

DOCUMENT CONTROL

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1. Executive Summary

The Enterprise AI Architecture Pattern (EAAP) defines an abstract enterprise design pattern for modular, governable, vendor-independent AI systems intended for controlled enterprise environments.

EAAP describes how enterprise AI capabilities can be structured as coordinated reasoning layers rather than isolated prompt implementations. The architecture separates domain intelligence, orchestration logic, knowledge grounding, and deployment control into clearly defined architectural layers.

EAAP is not a software product, executable implementation, or customer-specific deployment. It is a generalised architecture model intended to support future domain implementations, orchestration models, and deployment variants.

2. Purpose

This document defines the core architecture principles of EAAP and establishes the conceptual boundary between abstract architecture principles, domain-specific implementation layers, orchestration logic, and deployment models. It is the master reference for all EAAP documentation.

3. Scope

Included

- Architectural principles for modular enterprise AI systems
- Logical separation of reasoning layers
- Knowledge integration concepts
- Governance-oriented structural controls
- Deployment neutrality principles

Excluded

- Individual domain assistant implementations
- Enterprise-specific Proof-of-Concept material
- Executable software components
- Customer-specific orchestration rules

4. Architectural Position

This document belongs to Layer 3, the Architecture Pattern Layer. Layer 3 defines abstract architectural principles independent of any individual implementation or deployment. Layer 1 (Implementation) and Layer 2 (Orchestration) documents may reference this document but do not redefine its scope.

5. Core Architectural Principle

EAAP is based on the principle that the large language model itself does not constitute the enterprise system. The enterprise system emerges through the interaction of structured orchestration, domain reasoning specialisation, controlled knowledge retrieval, governance constraints, and deployment boundaries. The model acts as a reasoning component within a larger controlled architecture.

6. Core Components

EAAP consists of four principal architectural components, each operating as a distinct layer with controlled interfaces.

6.1 Domain Intelligence Layer

Provides domain-specific reasoning through bounded AI agents. Each agent operates within defined input, reasoning, evidence, output, confidence, and execution boundaries. Agents do not communicate directly with each other or with infrastructure systems.

6.2 Orchestration Layer

Provides supervisory reasoning and coordination when requests exceed single-domain scope. Determines primary domain authority, coordinates supporting domains, detects and resolves conflicts, assigns confidence, and enforces escalation. Advisory only unless connected to a controlled execution layer.

6.3 Knowledge Layer

Provides grounding for domain and orchestration reasoning through structured knowledge sources. Knowledge is classified by evidence class and consumed through controlled retrieval interfaces.

6.4 Governance Layer

Provides structural controls for traceability, auditability, escalation, and human review. Governance is an architectural component, not an optional overlay.

7. Layer Interaction Principle

- Each architectural layer operates independently through controlled interfaces.
- No layer implicitly transfers scope into another layer.
- Domain implementations may apply selected EAAP principles without representing the full architecture.
- Future orchestration models remain separate unless explicitly documented and agreed.
- The Governance Layer applies across all other layers without being owned by any single layer.

8. Deployment Principle

EAAP is deployment-neutral. The architecture supports air-gapped enterprise environments, sovereign enterprise deployments, and controlled hybrid deployments. No dependency on external APIs, cloud connectivity, or vendor-managed control planes is assumed.

9. Governance Principle

- EAAP assumes enterprise governance as a primary architectural requirement.
- Architectural outputs must support human review.
 - Evidence classification must be visible in all outputs.
 - Controlled escalation must be enforced at defined trigger points.
 - Separation between recommendation and execution must be maintained.
 - Audit trails must be produced for all governed actions.

10. Intellectual Position

EAAP constitutes an abstract architectural pattern independent of any Proof-of-Concept, domain implementation, or customer deployment. Future orchestration models, generalised reasoning structures, and deployment abstractions remain separate intellectual property unless expressly documented and commercially agreed.

11. Document Hierarchy

Document ID	Title	Layer
EAAP-CORE-001	Core Architecture Definition	Layer 3: Pattern
EAAP-ARCH-001	Domain Agent Model	Layer 2: Architecture
EAAP-ARCH-002	Orchestration Logic	Layer 2: Architecture
EAAP-ARCH-002A	Orchestration Prompt Control	Layer 2: Architecture
EAAP-ARCH-003	Orchestration Decision Model	Layer 2: Architecture
EAAP-ARCH-003A	Orchestration Decision Model (Production Design)	Layer 1: Implementation
EAAP-ARCH-004	Functional Architecture Model	Layer 2: Architecture
EAAP-ARCH-005	Functional Architecture HLD	Layer 1: Implementation
EAAP-ARCH-006	Deployment Model (Platform-Agnostic)	Layer 2: Architecture

Glossary

Term	Definition
EAAP	Enterprise AI Architecture Pattern. The abstract design pattern defined in this document.
Domain Agent	A bounded AI reasoning component dedicated to one enterprise domain.
Orchestration Layer	The supervisory reasoning layer that coordinates multiple domain agents.
Knowledge Layer	Structured knowledge sources grounding agent and orchestration reasoning.
Governance Layer	Structural controls ensuring traceability, auditability, and human oversight.
Evidence Class	Classification of the source supporting an output: Vendor-Supported, Framework-Supported, Best Practice, Heuristic, or Assumption.
Execution Boundary	The controlled limit separating AI reasoning from enterprise action.